

REMIND Lab - Non-Functional Requirements Specification

MVP v0.1 | Quality attributes and engineering constraints

1. Purpose

This document defines the initial non-functional quality baseline. It protects correctness, security and maintainability without inventing enterprise-scale constraints that have not been demonstrated.

2. Engineering principle

Complexity must be earned. Additional infrastructure is justified only by a real requirement, correctness need or measured constraint.

3. Non-functional requirements

ID	Quality	Area	Requirement	Priority
NFR-001	Security	Authentication	Protected operations require an authenticated principal.	MUST
NFR-002	Security	Authorization	Enforce role permissions plus object-level authority: ownership, assignment and category eligibility.	MUST
NFR-003	Security	Credentials	Locally stored credentials shall never be persisted in plaintext.	MUST
NFR-004	Integrity	Concurrency	Concurrent acceptance shall preserve at most one responsible Company per report.	MUST
NFR-005	Integrity	Atomicity	Lifecycle transitions and responsibility changes shall not leave partially applied state.	MUST
NFR-006	Auditability	Traceability	Retain sufficient actor references and timestamps to reconstruct report lifecycle.	MUST
NFR-007	Reliability	Invalid transitions	Invalid transitions fail deterministically without mutating persisted state.	MUST
NFR-008	Maintainability	Business isolation	Core business rules shall be testable without depending on HTTP or another delivery mechanism.	MUST
NFR-009	Maintainability	Automated tests	Core invariants, authority rules and state transitions shall have automated tests.	MUST
NFR-010	Performance	Measured performance	Do not invent an SLA before measurement; avoid obvious blocking/inefficient behavior in MVP workloads.	SHOULD
NFR-011	Scalability	Earned complexity	Distributed cache, brokers and service decomposition require demonstrated need or measured constraint.	MUST
NFR-012	Data	Referential integrity	Core relationships shall preserve referential integrity where supported by selected persistence.	MUST
NFR-013	Observability	Diagnostics	Failures and transition errors should be diagnosable through application logs without exposing secrets.	SHOULD
NFR-014	Portability	Configuration	Environment-specific configuration and secrets shall be externalized from source code.	MUST
NFR-015	Documentation	Traceability	Requirements, domain rules and significant architecture decisions shall be versioned with the repository.	MUST

4. Explicit non-requirements

- No AI/LLM integration.

- No Kafka or distributed message broker.
- No Redis or distributed cache.
- No microservices.
- No automatic geolocation dispatch.
- No invented requests-per-second target, uptime SLA or multi-region requirement before evidence exists.

5. Quality gates

- Automated tests cover core lifecycle and authority rules.
- Concurrent acceptance preserves a single responsible Company.
- Secrets and plaintext credentials are absent from source control.
- Invalid transitions preserve state.
- Implementation remains traceable to requirement IDs.